Analysis Tutorial 2

CRSALE010, Alex Cristaudo

Problem 4

(a) Let $\mathcal{A}$ be a σ-algebra

From $\Omega \in \mathcal{A}$, it follows that $\overline{\Omega} = \emptyset \in \mathcal{A}$

Now let $A, B \in \mathcal{A}$. It is clear that $A \cup B \in \mathcal{A}$.

Additionally, $A \setminus B = A \cap \overline{B}$. But $\overline{(A \cap \overline{B})} = \overline{A} \cup B \in \mathcal{A}$,

and so $A \setminus B \in \mathcal{A}$, thus $\mathcal{A}$ is a ring

(b) Let $\Omega = \{1,2,3\}$, and $\mathcal{A} = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$.

This is clearly a ring, but not a σ-algebra since $\Omega \notin \mathcal{A}$

(c) We must show $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{A} \Rightarrow \bigcup_{n=1}^{\infty} A_n$

$\Leftrightarrow \{A_n\}_{n\in\mathbb{N}} \subseteq A \Rightarrow \bigcap_{n=1}^{\infty} A_n$ (the other 2 properties are identical)

($\Rightarrow$) let $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{A}$, with $\mathcal{A}$ a σ-algebra.

Since $\overline{\bigcap_{n=1}^{\infty} A_n} = \bigcup_{n=1}^{\infty} \overline{A_n} \in \mathcal{A}$ since each

$\overline{A_n} \in \mathcal{A}$, we get $\bigcap_{n=1}^{\infty} A_n \in \mathcal{A}$.

($\Leftarrow$) let $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{A}$ with $\mathcal{A}$ a δ-algebra.

Similarly $\overline{\bigcup_{n=1}^{\infty} A_n} = \bigcap_{n=1}^{\infty} \overline{A_n} \in \mathcal{A}$ and so $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$.

So a family of sets $\mathcal{A}$ is a σ-algebra

$\Leftrightarrow \mathcal{A}$ is a δ-algebra

Problem 5

① Since $\emptyset$ is at most countable, $\overline{\emptyset} = \Omega \in \mathcal{A}$

② Clearly if $A \in \mathcal{A}$, $\overline{A} \in \mathcal{A}$ (as then either

$\overline{A}$ or A is at most countable

③ Let $\{A_n\}_{n\in\mathbb{N}} \subseteq \mathcal{A}$. Then ei A_n or $\overline{A_n}$ is

at most countable. for each $n \in \mathbb{N}$.

If, for any $N \in \mathbb{N}$, $\overline{A_N}$ is at most countable, then $\overline{\bigcup_{n=1}^{\infty} A_n} = \bigcap_{n=1}^{\infty} \overline{A_n} \subseteq \overline{A_N}$ is at most countable so $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$. On the other hand, if for every $N \in \mathbb{N}$, $\overline{A_N}$ is uncountable, then A_N is at most countable. But then $\bigcup_{n=1}^{\infty} A_n$ is at most countable, so $\bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$. So $\mathcal{A}$ is a σ-algebra

Problem 6

Let $A = \{(x,y) : x, y \in \mathbb{R}\}$
$B = \{(x,y) : x, y \in \mathbb{Q}\}$
$C = \{(-\infty, x] : x \in \mathbb{R}\}$
$D = \{(-\infty, x] : x \in \mathbb{Q}\}$

Note $B \subseteq A$. Let $x, y \in \mathbb{R}$. For each $n \in \mathbb{N}$, we can find $q_n^{(1)}, q_n^{(2)} \in \mathbb{Q}$ such that:
$0 \le q_n^{(1)} - x < \frac{1}{n}$
$0 \le y - q_n^{(2)} < \frac{1}{n}$
It follows that $\bigcup_{n=1}^{\infty} (q_n^{(1)}, q_n^{(2)}) = (x,y) \in \sigma(B)$.
Thus $\sigma(A) = \sigma(B)$
We can do a similar argument to show $D \subseteq C \subseteq \sigma(D) \Rightarrow \sigma(C) = \sigma(D)$
(taking $x \in \mathbb{R}$, $\exists\, q_n \in \mathbb{Q}$, $0 \le x - q_n < \frac{1}{n}$)

Now let $x \in \mathbb{R}$. Then $(-\infty, x] = (-\infty, x) \cup \big(\underbrace{\bigcap_{n=1}^{\infty} (x - \tfrac{1}{n}, x + \tfrac{1}{n})}_{\in \sigma(A)}\big) \in \sigma(A)$. So $C \subseteq \sigma(A) \Rightarrow \sigma(C) \subseteq \sigma(A)$

Also, for $x, y \in \mathbb{R}$, $(x,y) = \left(\bigcup_{n=1}^{\infty} (-\infty, y - \tfrac{1}{n}]\right) \cap \overline{(-\infty, x]} \in \sigma(C)$, so $A \subseteq \sigma(C)$ thus $\sigma(A) \subseteq \sigma(C)$ so $\sigma(A) = \sigma(C)$ and finally $\sigma(A) = \sigma(B) = \sigma(C) = \sigma(D)$